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Review

Long-Duration Energy Storage: A Critical Enabler for Renewable Integration and Decarbonization

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Abstract: This paper focuses on the critical role of long-duration energy storage (LDES) technologies in facilitating renewable energy integration and achieving carbon neutrality. It presents a systematic review of four primary categories: mechanical energy storage, chemical energy storage, electrochemical energy storage, and thermal energy storage. The study begins by analyzing the technical advantages and geographical constraints of pumped hydro energy storage (PHES) and compressed air energy storage (CAES) in high-capacity applications. It then explores the potential of hydrogen and synthetic fuels for long-duration clean energy storage. The section on electrochemical energy storage highlights the high energy density and flexible scalability of lithium-ion batteries and redox flow batteries. Finally, the paper evaluates innovative advancements in large-scale thermal energy storage technologies, including sensible heat storage, latent heat storage, and thermochemical heat storage. By comparing the performance metrics, application scenarios, and development prospects of various energy storage technologies, this work provides theoretical support and practical insights for maximizing renewable energy utilization and driving the sustainable transformation of global energy systems.

Keywords: long-duration energy storage; carbon neutrality; large scale; renewable energy

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1. Introduction

Since the 21st century, global climate change has garnered widespread attention, prompting countries to implement measures to reduce greenhouse gas emissions, particularly CO^2. The Paris Agreement explicitly proposes achieving carbon neutrality by the second half of this century. Carbon neutrality refers to balancing CO_2 emissions generated by human activities with the amount removed from the atmosphere through sequestration or offsetting, ultimately achieving a net-zero emissions state [1]. A crucial approach to achieving this goal is the large-scale adoption of renewable energy to reduce CO_2 emissions from conventional fossil fuels. According to the International Renewable Energy Agency (IRENA), the share of renewables in global electricity generation must increase from approximately 25% in 2020 to 60% by 2030 to meet net-zero targets [2].

However, as the share of variable renewable energy (VRE) in power systems continues to grow, its intermittency, volatility, and limited dispatchability pose challenges to grid stability. Reliable grid operation depends on precise supply–demand matching, but the uncertainty of VRE generation can lead to imbalances across different time scales, reducing grid reliability.

Energy storage technologies are critical to addressing these challenges. They can store surplus electricity during periods of low demand and release it during peak demand, balancing grid loads and minimizing the impact of VRE on the grid. Based on storage duration, energy storage technologies are classified into short-duration energy storage and LDES. While there is no universally accepted definition of LDES, it is generally characterized by storage durations of at least several days, with some technologies capable of seasonal energy storage [1,3,4].

Historically, research has primarily focused on short-duration energy storage technologies. However, recent studies indicate that developing LDES technologies is essential for integrating high shares of renewables into power grids. Supporting this shift, the number of publications on long-duration energy storage technologies has increased significantly over the past decade. For instance:

- Research on mechanical energy storage has grown from 345 articles in 2015 to 780 articles in 2024, representing a 125% increase.
- Studies on chemical energy storage have surged from 6717 articles in 2015 to 16,619 articles in 2024, marking a 147% rise.
- Similarly, research on electrochemical energy storage has doubled, from 7229 articles in 2015 to 15686 articles in 2024.
- Finally, thermal energy storage publications have grown from 969 articles in 2015 to 2539 articles in 2024, reflecting a 162% increase.

LDES technologies are categorized into four major types based on their storage mechanisms:

1. Mechanical energy storage, including pumped hydro energy storage (PHES) and compressed air energy storage (CAES).
2. Chemical energy storage, encompassing hydrogen storage and synthetic fuel storage.
3. Electrochemical energy storage, represented by redox flow batteries and lithium-ion batteries.
4. Thermal energy storage, comprising sensible heat storage (STS), latent heat storage (LTS), and thermochemical heat storage (THS).

Each energy storage technology exhibits unique characteristics in terms of storage form, cost, efficiency, capacity, and scalability, making them suitable for different application scenarios. Figure 1 below illustrates different types of LDES technologies.

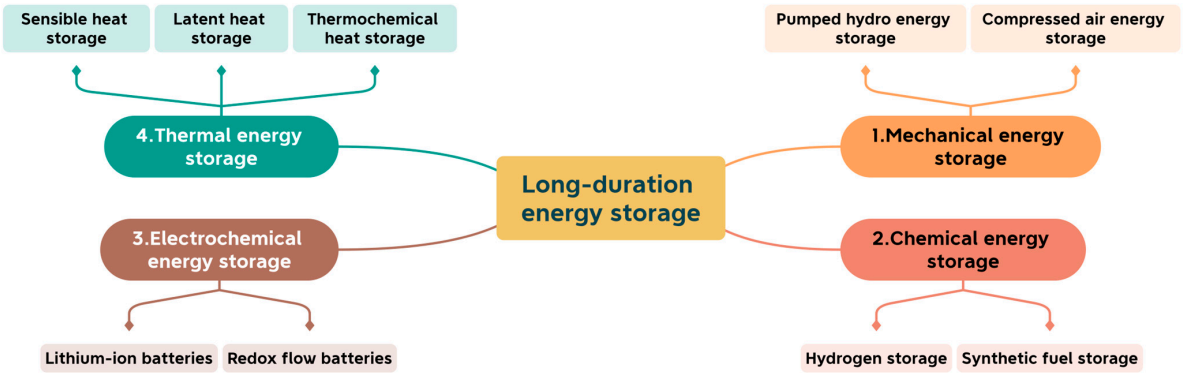


Figure 1. Types of LDES.

This paper systematically reviews the current state of LDES technologies, discussing the fundamental principles, performance metrics, and application scenarios of mechanical energy storage, chemical energy storage, electrochemical energy storage, and thermal energy storage. By comparing energy density, conversion efficiency, and environmental adaptability, this study analyzes the advantages and disadvantages of various storage

technologies. Finally, it provides future perspectives on LDES development and offers recommendations for advancing research, serving as a theoretical foundation and practical guide for future endeavors.

2. Mechanical Energy Storage

2.1. Pumped Hydro Energy Storage

2.1.1. Overview

PHES is a mature energy storage technology that converts electrical energy into gravitational potential energy for storage and subsequent release. As the penetration of intermittent renewable energy sources like wind and solar power in the grid continues to rise, large-scale energy storage technologies have become essential for maintaining grid balance and stability. As an indirect utilization of hydropower, PHES offers high safety, stability, and low operational costs, playing a vital role in renewable energy integration and grid peak shaving [5].

The fundamental principle of PHES involves two main stages: during periods of low electricity demand, surplus electricity is used to power pumps that transfer water from a lower reservoir to an upper reservoir, storing energy as gravitational potential. During periods of high electricity demand, the stored water is released to drive turbines, converting gravitational potential energy back into electricity and supplying it to the grid to ensure load stability. PHES systems integrated with solar or wind power utilize photovoltaic or wind energy to power pumps during low demand and combine upper reservoir water with renewable sources to supply power to the grid during high demand, achieving efficient energy utilization [6].

To further enhance the understanding of PHES functionality, Figure 2 illustrates a schematic diagram of its operation. Key components of the system, such as the surge chamber and transformer, play crucial roles in its performance. The surge chamber acts as a buffer to regulate water flow, preventing hydraulic shocks during rapid changes in operation, thereby ensuring system stability and safety. The transformer, on the other hand, facilitates the conversion of electricity between different voltage levels during both pumping and generation stages, optimizing energy transfer efficiency and minimizing losses. These components, coupled with the integration of renewable energy sources, make PHES an indispensable solution for balancing supply and demand in modern power grids.

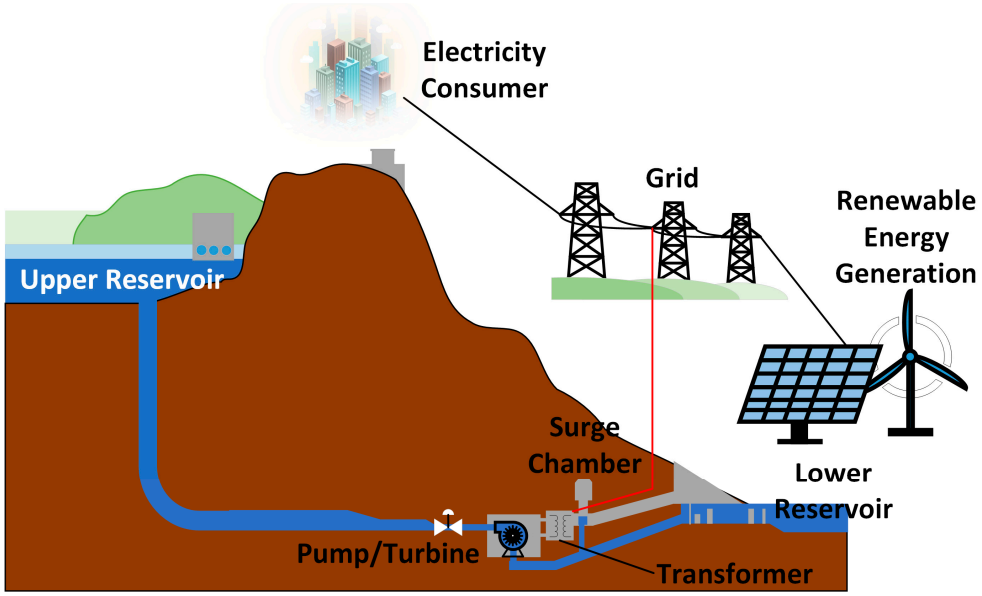


Figure 2. Schematic diagram of PHES.

2.1.2. Advantages and Challenges

PHES currently accounts for approximately 96% of global energy storage capacity and 99% of storage volume, making it the largest deployed energy storage technology worldwide [7]. Its significant advantages include large capacity, long lifespan, and high reliability, making it suitable for large-scale grid energy storage. Additionally, in regions with abundant rainfall, PHES systems can serve flood control functions through reservoir structures, enhancing their diversified application value [8].

Despite its many advantages, PHES faces certain geographical limitations. For instance, it requires sufficient water resources and substantial elevation differences, which restrict its applicability in arid or low-altitude regions [9,10]. Although research on ultra-low-head PHES is underway, breakthroughs in efficiency and economic viability have yet to be achieved [11].

2.1.3. Outlook

Future research on PHES will focus on the following areas: first, improving cycle efficiency. Current systems achieve a power generation efficiency of approximately 90%, meaning about 10% of energy is lost in a full charge–discharge cycle. Enhancing efficiency will significantly reduce energy waste, providing greater support for achieving carbon neutrality goals. Second, increasing operational flexibility by exploring applications in novel scenarios such as high-rise buildings, artificial lakes, and abandoned mines to reduce dependence on topography and water resources [12–14]. Third, optimizing designs and adopting intelligent management to minimize environmental impacts and enhance sustainability. Examples include integrating real-time control algorithms for water flow management and utilizing pre-existing reservoirs to lower construction costs and land-use impacts. Additionally, employing advanced economic models such as levelized cost of storage (LCOS) assessments has proven effective in identifying cost-efficient configurations [3,15].

As a core technology in the energy storage domain, PHES will continue to play a critical role in future grid balancing, renewable energy integration, and the global energy transition.

2.2. Compressed Air Energy Storage

2.2.1. Overview

CAES can be traced back to the 1940s [16]. Initially, this technology was primarily used to balance fluctuations between peak and off-peak electricity demand [17]. With ongoing advancements, CAES has gradually demonstrated its unique advantages in the energy storage sector [18]. The basic principle of CAES involves storing energy as elastic potential energy in compressed air at pressures up to 8 MPa. The power generation process is similar to that of conventional gas turbines.

A CAES system consists of six key components: an electric motor and generator (operating alternately with the compressor or turbine), air compressor, high- and low-pressure turbines, combustion turbines, auxiliary equipment, and air storage containers [19]. The operational process includes two stages: during the storage phase, electricity drives the compressor to compress air, which is then stored in containers; during the discharge phase, the stored air is released, mixed with fuel, and combusted to drive a turbine for power generation. Air storage containers can be artificial tanks, but in most cases, natural underground cavities such as caverns, abandoned mines, or aquifers are preferred to reduce costs, which are typically one-tenth of artificial facilities [20].

2.2.2. Types and Technical Characteristics

CAES is advantageous for its large storage capacity, high pressure, and low construction cost, making it particularly suitable for smoothing the output of intermittent renewable

energy and enabling LDES applications [21]. With ongoing research, CAES technology has evolved into three primary types: diabatic CAES (D-CAES), adiabatic CAES (A-CAES), and isothermal CAES (I-CAES).

D-CAES is the earliest form of CAES technology. During air compression, significant heat is generated, and without proper cooling, storage containers could be damaged due to high temperatures, affecting the system's lifespan. To address this, D-CAES employs multiple coolers to transfer the heat to the external environment, reducing the air temperature to near ambient levels and ensuring operational safety. However, this cooling method leads to complete heat loss, resulting in an energy efficiency of only 40–50% [21]. Additionally, as air must be preheated by mixing with fuel before power generation, D-CAES requires supplemental fossil fuel heating, thereby increasing carbon emissions.

A-CAES addresses the energy loss issue of D-CAES by recovering and storing the heat generated during compression. In A-CAES systems, the air temperature is maintained between 30 °C and 50 °C [22], ensuring safe storage and providing preheated air for subsequent power generation, thereby reducing the need for external heating. A-CAES systems achieve energy efficiencies of 55–75%, and potentially higher depending on the thermal storage method. However, A-CAES requires extended operation to reach optimal efficiency, making it unsuitable for short-duration, high-frequency storage applications [23].

I-CAES maintains a constant temperature during compression, minimizing heat loss while reducing thermal stress on equipment. Without the need for additional heat exchangers, I-CAES systems feature simpler designs and higher energy density. The ideal energy efficiency of I-CAES can exceed 90% [24], making it highly promising for LDES applications [25,26]. However, I-CAES currently relies on piston-based compression systems, resulting in lower power output levels [20]. Additionally, achieving true isothermal compression in practical operations remains a significant technical challenge.

A comparison of the three CAES technologies is shown in Table 1.

Table 1. Comparison of CAES.

CAES Technology	Advantages	Disadvantages
D-CAES	Mature technology Simple system structure	Requires additional fossil fuel heating Significant energy loss during compression
A-CAES	Minimal energy loss during compression Low storage temperature High system efficiency	Slow system response
I-CAES	Negligible energy loss during compression High energy density Very high system efficiency	Complex system structureLow power output level

2.2.3. Outlook

Overall, CAES is a highly competitive LDES technology, characterized by large energy capacity, long operational lifespan, and minimal energy loss over extended storage periods. While PHES demonstrates greater maturity in terms of application range and cycle efficiency within the mechanical energy storage domain, CAES offers a distinct advantage: its flexibility to be deployed in flat terrains and mountainous regions without dependence on specific topography, and with limited susceptibility to climatic variations.

Current research focuses on enhancing CAES energy efficiency, improving response speed, optimizing discharge depth, and achieving seamless integration with renewable energy grids. These advancements aim to further drive CAES toward achieving the goal of zero-carbon emissions. For instance, recent studies have explored optimizing

compressor designs and integrating thermal energy storage systems to recover and utilize heat generated during compression, significantly improving overall system efficiency [20].

3. Chemical Energy Storage

3.1. Hydrogen Energy Storage

3.1.1. Overview

Hydrogen energy is a representative form of long-duration chemical energy storage, capable of storing energy in the form of hydrogen gas for months or even longer. Hydrogen can be produced through various methods. The traditional approach, catalytic reforming of fossil fuels, produces "grey hydrogen." However, this method heavily relies on fossil fuels, making it incompatible with low-carbon development goals. In contrast, hydrogen production via water electrolysis, which can directly utilize intermittent renewable energy, offers high flexibility and is considered a key technological pathway for achieving LDES. Hydrogen produced through this method is referred to as "green hydrogen." Additionally, technologies like solar thermochemical decomposition can directly produce hydrogen. However, due to its technological maturity and compatibility with renewable energy, water electrolysis has become the focus of current research and applications [27,28]. Figure 3 illustrates the hydrogen storage.

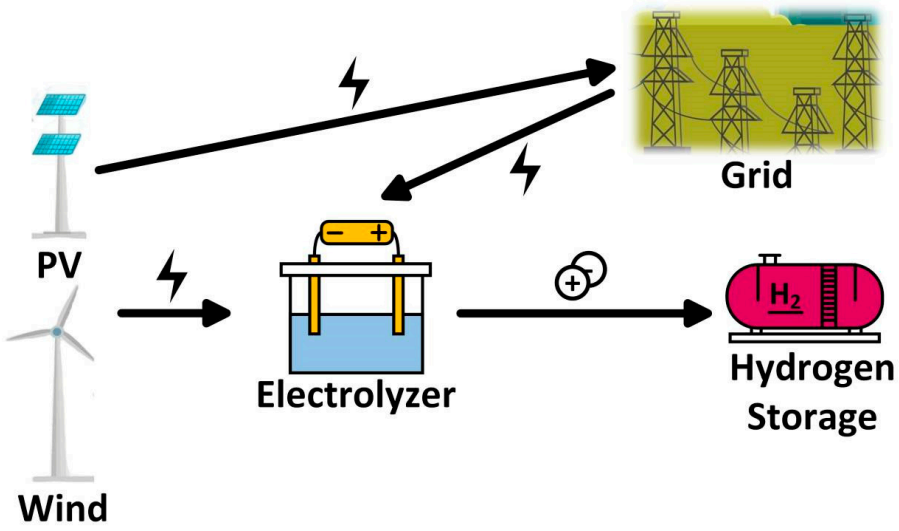


Figure 3. Schematic diagram of hydrogen storage.

Despite these advantages, hydrogen energy presents significant safety risks. As a highly flammable gas, hydrogen can form explosive mixtures with air under certain conditions, necessitating meticulous handling and storage measures. The processes of transportation and refueling introduce additional risks, such as fire or explosion, emphasizing the critical need for specialized storage equipment and the implementation of stringent safety protocols to mitigate potential hazards.

Beyond providing a clean energy supply, hydrogen energy offers diverse additional applications in the LDES field. Stored hydrogen can be utilized as heating and cooling energy [29], as a clean heat source and reducing agent replacing coke in steel production [30], and in the chemical industry for synthesizing ammonia and methanol [31]. Furthermore, hydrogen is an essential energy source in the aerospace sector [32].

3.1.2. Types and Technical Characteristics

Current hydrogen storage methods primarily include gaseous, liquid, and solid-state storage, each differing significantly in energy density and applicability.

Gaseous hydrogen storage relies on high-pressure containers, categorized by pressure levels: type I (<20 MPa), type II (<20 MPa), type III (20–30 MPa), type IV (30–45 MPa), and type V (>45 MPa). Its advantages include operational convenience and fast refueling. However, it has notable drawbacks, such as low energy density (approximately 40 kg/m^3 at 70 MPa), high leakage risk, and significant energy consumption during refueling (approximately 3–4 kWh/kg) [33,34].

Liquid hydrogen storage involves cooling hydrogen to −253 °C for liquefaction. This method achieves approximately twice the energy density of gaseous hydrogen and requires smaller storage volumes, making it suitable for large-scale energy storage. However, it faces challenges such as high energy consumption, expensive storage tanks, and boil-off losses (approximately 3% per day), which limit its economic viability for LDES [35]. Additionally, another approach is liquid organic hydrogen carriers (LOHCs), which absorb or adsorb hydrogen into organic compounds, avoiding the high energy costs of cryogenic liquefaction. However, LOHCs have limitations, including low power density, slow absorption and release rates, and the risk of carrier degradation during cycling [36,37].

Solid-state hydrogen storage utilizes materials such as activated carbon and metal hydrides for physical or chemical adsorption of hydrogen. This method offers the highest energy density and is considered the ideal approach for future long-duration hydrogen storage. Nonetheless, all hydrogen storage methods face significant challenges, including low energy density, high storage costs, and safety concerns [38].

A comparison of hydrogen storage methods is shown in Table 2.

Table 2. Comparison of hydrogen storage methods.

Hydrogen Storage Method	Advantages	Disadvantages
Gaseous hydrogen storage	Fast refueling Convenient operation Relatively low cost	Low energy density High leakage risk High energy consumption during refueling
Liquid hydrogen storage	Higher energy density	Requires cryogenic storage (−253 °C) High losses over long-duration storage Significant energy consumption during storage
Solid-state hydrogen storage	Highest energy density	Technology is still in the development stage

3.1.3. Outlook

The future development of hydrogen energy storage requires breakthroughs in several key areas. In hydrogen production, the energy consumption and efficiency of water electrolysis remain critical bottlenecks that need to be addressed. Future research should focus on developing novel catalytic materials and more efficient electrolysis technologies to reduce energy consumption and improve hydrogen production efficiency. Additionally, innovative solutions such as seawater electrolysis, which is suitable for water-scarce regions, should be explored to reduce reliance on freshwater resources.

In hydrogen storage, increasing energy density is a primary task. Advancements in high-density hydrogen storage materials, such as more efficient metal hydrides and LOHCs, alongside improvements in existing gaseous and liquid hydrogen storage technologies, can further reduce storage costs and enhance system safety. Moreover, the development of solid-state hydrogen storage technologies that offer high capacity and excellent cycle stability, coupled with the optimization of material fabrication processes and recycling mechanisms, will provide competitive solutions for LDES applications.

Through comprehensive innovation in hydrogen production and storage technologies, hydrogen energy storage is poised to play a more significant role in future energy

systems, supporting the global energy transition and the achievement of sustainable development goals.

3.2. Synthetic Fuel Energy Storage

3.2.1. Overview

Synthetic fuels are a promising alternative to traditional fossil fuels. Conventional fuel synthesis processes often have severe environmental impacts. However, with the rapid development of renewable energy and the continuous reduction in its costs, the synthesis of fuels using renewable energy has emerged as a viable solution for LDES [39]. Renewable synthetic fuels are produced by hydrogenating CO_2 or N_2 to create hydrocarbon fuels (e.g., methane and methanol) or nitrogen-hydrogen compounds (e.g., ammonia). Due to their high energy density and excellent chemical stability, these fuels are particularly well-suited for LDES. Moreover, since their production is powered by renewable energy, they are often referred to as "E-fuels" [40].

Common renewable synthetic fuels include methane (E-methane), methanol (E-methanol), and ammonia (E-ammonia). These synthetic fuels are produced using catalysts under appropriate pressure and temperature conditions. The key chemical reaction equations are as follows (1)–(3).

$$CO_2 + 4H_2 \xrightarrow[\text{catalyst}]{\substack{250\text{–}400\ ^\circ\text{C} \\ 5\text{–}50\text{bar}}} CH_4 + 2H_2O \qquad \Delta H = -165\text{KJ/mol} \tag{1}$$

$$CO_2 + 3H_2 \xrightarrow[\text{catalyst}]{\substack{200\text{–}300\ ^\circ\text{C} \\ 50\text{–}100\text{bar}}} CH_3OH + H_2O \quad \Delta H = -49.2\text{KJ/mol} \tag{2}$$

$$CO_2 + 3H_2 \xrightarrow[\text{catalyst}]{\substack{200\text{–}300\ ^\circ\text{C} \\ 50\text{–}100\text{bar}}} CH_3OH + H_2O \quad \Delta H = -49.2\text{KJ/mol} \tag{3}$$

3.2.2. Advantages and Challenges

Renewable synthetic fuels have garnered significant attention for their potential in LDES and their contribution to carbon neutrality. Carbon-based renewable synthetic fuels can utilize atmospheric CO_2 as a carbon source, allowing them to serve not only as energy storage media but also as a means to capture and recycle atmospheric carbon dioxide during storage and release processes. Additionally, biomass can also be employed as an alternative carbon source for their production, as illustrated in Figure 4, further enhancing their versatility and sustainability [41]. In terms of applications, renewable synthetic fuels share similarities with hydrogen in their versatility. They can be used as fuels for internal combustion engines and gas turbines in industrial applications [39], as chemical feedstocks [42], or as hydrogen carriers in fuel cells [43].

Despite their notable advantages in LDES, renewable synthetic fuels face several challenges. The most prominent issue is their high cost, encompassing expenses for infrastructure construction and installation, electricity supply, safety management, transportation, and distribution. Although the IRENA's latest report highlights the declining costs of renewable energy technologies such as wind and solar power, making renewable electricity more affordable [44], associated auxiliary costs remain significant. Additionally, the high cost of hydrogen production is another limiting factor. As discussed in the hydrogen

energy section, water electrolysis technology still requires further optimization to improve economic feasibility.

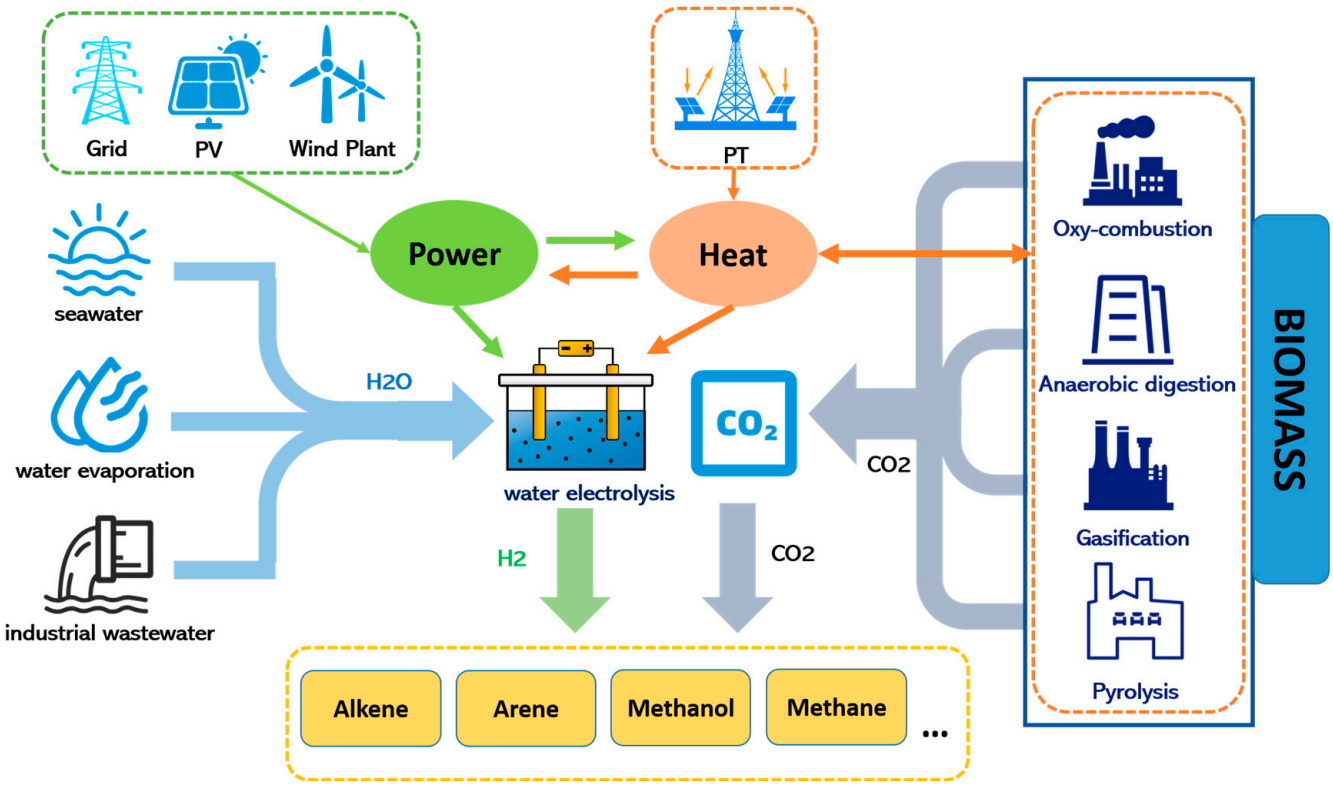


Figure 4. Production of synthetic fuels using biomass as a carbon source [42].

Another critical limitation lies in the efficiency gap of production pathways. The synthesis processes for renewable synthetic fuels are less efficient compared to conventional synthetic fuel production and have significantly higher production costs than fossil-based synthetic fuels [39]. Consequently, reducing production costs through process improvement and technological breakthroughs is imperative for future development.

In summary, renewable synthetic fuels, as high-quality energy sources, are not only suitable for LDES but also find broad applicability across various sectors. With continued technological advancements, particularly in cost reduction and process optimization, their future development prospects are highly promising.

4. Electrochemical Energy Storage

Electrochemical energy storage relies on chemical reactions within batteries to store and release energy. Among various energy storage technologies, it has garnered significant attention due to its high energy density, flexible scalability, and environmental friendliness. Electrochemical energy storage plays a critical role in LDES, particularly in renewable energy integration and grid regulation applications. With stable performance and rapid response capabilities, it significantly enhances the reliability of energy systems. Notably, lithium-ion batteries (LIBs) and redox flow batteries (RFBs), as representative electrochemical energy storage technologies, have demonstrated broad application prospects in LDES. The following sections provide detailed discussions of these two technologies.

4.1. Lithium-Ion Battery

LIBs have become one of the most widely applied technologies in electrochemical energy storage due to their high energy density, long cycle life, and technological maturity.

Their rapid response capability, high energy efficiency, and high energy density have made them central to renewable energy and grid integration efforts. Current LIBs typically consist of an anode, cathode, separator, and liquid electrolyte. Energy storage and release are achieved through reversible intercalation/de-intercalation reactions between the anode and cathode. LIBs boast an energy density of up to 265 Wh/kg and an efficiency of 97%, significantly surpassing mechanical and chemical energy storage technologies. Additionally, their low self-discharge rate (as low as 0.03%) provides a distinct advantage in LDES applications [45].

Despite their promising prospects in LDES, LIBs face several challenges in practical applications. First, due to the large scale of renewable energy generation, LIB systems often consist of thousands of battery modules. Managing consistency across such systems is highly challenging, as individual batteries, if overcharged, may experience irreversible changes in the electrodes, accelerating degradation. Additionally, interactions among battery components during charge and discharge cycles can lead to electrode material deterioration, further compromising battery performance and safety [46,47]. The aging and degradation of LIBs involve complex physicochemical mechanisms, necessitating deeper research to optimize battery performance.

Safety is another major concern for LIBs. Traditional LIBs with liquid electrolytes and plastic separators generate heat during high-power charge and discharge cycles, potentially leading to fires or even explosions if mishandled. Recent incidents of storage station fires highlight these safety limitations [48]. To enhance performance and safety, solid-state LIBs are widely regarded as the next-generation solution. Using solid electrolytes, these batteries theoretically achieve an energy density of up to 500 Wh/kg, with significantly improved lifespan and safety compared to traditional LIBs [49,50]. Recent studies have shown that single-ion polymer electrolytes, such as PAF-322-Li, can significantly reduce lithium dendrite formation and improve ionic conductivity, achieving stable cycling even at −20 °C [51]. Furthermore, the introduction of solid electrolytes enables the implementation of bipolar designs within sealed battery packs, increasing energy capacity and better meeting the demands of large-scale renewable energy storage [52]. However, solid-state LIBs exhibit reduced performance at low temperatures and higher manufacturing costs, requiring further research to improve their economic and practical feasibility.

4.2. Redox Flow Battery

RFBs are specialized electrochemical energy storage systems. Unlike other types of batteries, their energy is stored in external tanks containing electrolyte solutions, and energy is released through reversible redox reactions within the battery [53]. This technology features a modular design, allowing energy capacity and power to be scaled independently, making it highly flexible for large-scale energy storage applications. The fundamental structure of an RFB includes positive and negative electrodes, an ion exchange membrane, and two storage tanks for housing the anolyte and catholyte solutions, respectively [54].

During charging, the anolyte and catholyte are pumped into the battery stack, where redox reactions are driven by an external renewable energy source. Oxidation occurs in the anolyte, while reduction occurs in the catholyte. The ion exchange membrane effectively prevents the mixing of electrolytes while allowing ions to pass through, maintaining the electrochemical balance of the system [55]. Once charging is complete, the electrolyte solutions are returned to the storage tanks. During discharge, reverse redox reactions occur, generating electricity as electrons flow through an external circuit. Figure 5 illustrates the working principle of an all-vanadium RFB.

Based on the electrolyte solutions used, RFBs can be classified into different types, including all-vanadium flow batteries, zinc-bromine flow batteries, and iron-chromium

flow batteries, each with unique characteristics in terms of energy density, cost, and performance [56]. Among these, all-vanadium flow batteries are particularly notable for their distinctive chemical properties and LDES capabilities. Their electrolyte contains vanadium ions in four oxidation states: +4 and +5 (VO^{2+} and $VO_2{}^+$ in the anolyte) and +2 and +3 (V^{2+} and V^{3+} in the catholyte). Energy storage and release are achieved through the redox reactions of vanadium ions in the electrolyte.

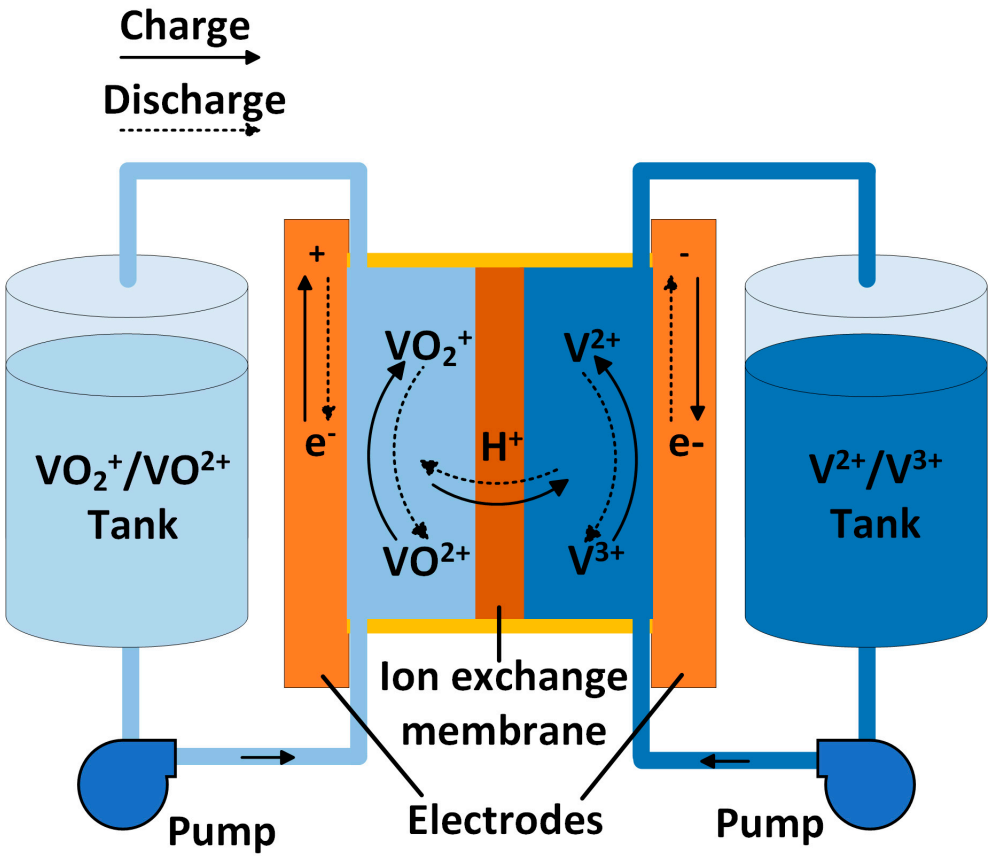


Figure 5. Working principle of an all-vanadium RFB.

Zinc-bromine flow batteries are another option for LDES, using zinc bromide solutions as the electrolyte. During charging, zinc is deposited as a solid on the negative electrode; during discharge, zinc releases electrons and re-enters the solution. Since active materials partially deposit on the electrodes, zinc-bromine flow batteries are classified as hybrid flow batteries, whereas systems like all-vanadium flow batteries, in which all active materials are fully dissolved in the electrolyte, are known as pure flow batteries [53].

RFBs demonstrate several advantages for LDES. Their energy is stored in external electrolyte tanks, allowing capacity to be increased by expanding tank size without modifying the battery stack structure. The separation of anolyte and catholyte in independent tanks ensures electrolyte stability prior to discharge, enabling storage durations ranging from several days to months, with efficiencies between 65% and 80% [1]. The modular design and low heat accumulation contribute to the high safety of RFB systems. Although the initial construction cost is relatively high, expanding capacity requires only adding tanks and electrolytes, reducing per-unit storage costs as the system scales. Furthermore, since electrochemical reactions occur primarily in the electrolyte and the electrodes serve only as carriers, performance degradation is minimized, resulting in cycle lives exceeding 15,000 cycles with exceptional stability.

In summary, RFBs demonstrate significant advantages in meeting large-scale energy storage demands due to their scalable capacity and high stability. However, their future development faces challenges related to electrolyte supply and cost. Optimizing electrolyte formulations and electrode materials will be key research priorities to further enhance their performance and economic viability.

5. Thermal Energy Storage

Thermal energy is one of the most suitable energy forms for long-duration storage due to its significant seasonal variability in nature. In summer, thermal energy is often abundant, especially during solar energy utilization, where excess heat typically needs to be dissipated. In contrast, thermal energy becomes scarce in winter, and similar seasonal demands are observed for "cold energy." The concept of thermal energy storage can be traced back to ancient China. For instance, during winter, ice blocks were stored in cellars, where the dryness, ventilation, and consistently low temperatures helped preserve the ice until summer for later use. This simple storage method reflects the basic principles of thermal energy storage.

With advancements in science and technology, modern thermal energy storage technologies have made significant progress. Contemporary thermal energy storage systems generally consist of storage media and devices used to inject or extract heat from the media. Based on the specific form of stored thermal energy, thermal energy storage can be categorized into three main types: sensible heat storage, latent heat storage, and thermochemical heat storage.

Sensible heat storage is the most traditional and widely applied method, directly storing thermal energy through temperature changes in materials. Latent heat storage relies on materials absorbing or releasing heat during phase transitions to store energy. Thermochemical heat storage stores and releases thermal energy through reversible chemical reactions [57].

The following subsections delve into the applications of sensible heat storage, latent heat storage, and thermochemical heat storage in LDES. They analyze their working principles, technical advantages, and future development prospects.

5.1. Sensible Heat Storage

5.1.1. Overview

Sensible heat storage is a critical component of LDES technologies, enabling direct thermal energy storage using media such as water, soil, rocks, or molten salts. It excels in large-scale, seasonal heat storage applications. Current sensible heat storage technologies include aquifer thermal energy storage (ATES), tank thermal energy storage (TTES), pit thermal energy storage (PTES), borehole thermal energy storage (BTES), and molten salt energy storage.

Each technology has unique operational mechanisms and application scenarios. For instance, ATES utilizes underground aquifers for heat storage, offering low costs and large storage capacity but facing significant geographical limitations. TTES and PTES, on the other hand, employ above-ground tanks or excavated pits for heat storage, providing high flexibility and broad applicability. BTES uses deep boreholes to store heat in underground soil or rocks, delivering high thermal stability and efficiency. Meanwhile, molten salt energy storage demonstrates immense potential for long-duration heat storage due to its large storage capacity and high efficiency.

The application of these technologies significantly enhances the utilization of renewable energy, mitigates energy fluctuations, and improves the reliability of energy systems.

5.1.2. Types and Technical Characteristics

Aquifer Thermal Energy Storage:

ATES is a type of sensible heat storage technology that uses underground aquifers as the storage medium. Its basic principle involves pumping groundwater, heating or cooling it via a heat exchanger, and then storing the thermal energy in the aquifer. The stored water temperature is typically maintained at around 25 °C [58]. Depending on the storage configuration, ATES can be classified into two types: well doublet systems, where warm and cold water are alternately stored across multiple horizontal wells, and monowell systems, which use vertical stratification to separate warm and cold water layers [59].

ATES typically achieves a storage efficiency between 67.5% and 87%, with potential improvements as the storage scale increases [60]. This technology is widely recognized for its low cost and large storage capacity. However, its applicability is significantly constrained by geographical location, and its operation can pose environmental risks, such as groundwater contamination, limiting its widespread adoption.

Tank/Pit Thermal Energy Storage:

Similar to ATES, TTES also uses water as the storage medium. However, in TTES, the water is stored in above-ground tanks, making it independent of geological conditions. High-quality insulation materials are used to enhance storage temperature and efficiency [61]. Additionally, tanks can be partially or fully embedded underground to further improve thermal performance. This design has led to the development of PTES, which involves excavating storage spaces, lining them with insulation materials, and using the excavated soil to raise embankments, thereby increasing storage capacity [62]. TTES/PTES systems generally achieve a storage efficiency of 45–65% [62].

A significant advantage of PTES over TTES is its ability to use liquid–solid phase change materials, such as water–gravel mixtures. Although such mixtures have lower heat capacity compared to pure water, requiring larger storage volumes, the solid phase material can often be sourced from the excavated soil itself, eliminating the need for expensive storage tanks and significantly reducing construction costs [63,64].

Borehole Thermal Energy Storage:

BTES stores thermal energy in underground soil or rocks through boreholes, which typically range from 30 to 200 m in depth [65]. Heat is extracted using heat pumps for energy supply. The stable temperatures found at greater depths allow BTES systems to minimize heat losses [66], making them ideal for long-duration heat storage. Installing insulation materials underground can further reduce heat loss and improve efficiency. The storage temperature of BTES depends largely on the heat source used. Solar thermal collectors are commonly employed, achieving storage temperatures of up to 90 °C [67]. BTES systems can also integrate waste heat from industrial processes or geothermal systems to enhance energy utilization efficiency.

BTES efficiency is influenced by various factors, including design, material properties, geological conditions, and operating parameters. While initial storage efficiency is low during the first year of operation, it can improve to 40–60% after several years [68]. However, the adoption of BTES faces challenges such as the high construction costs of deep drilling and the sensitivity of storage efficiency to external conditions.

Molten Salt Energy Storage:

Molten salt energy storage stores thermal energy using liquid molten salts at high temperatures, primarily to address the intermittency of concentrating solar power plants. With the growing demand for LDES, molten salt energy storage has demonstrated significant potential due to its high storage capacity and efficiency [69]. Commonly used molten salts include nitrates ($NaNO_3/KNO_3$), with maximum storage temperatures reaching approximately 565 °C [70]. Chlorides ($MgCl_2$/NaCl/KCl) can achieve even higher

storage temperatures, enabling greater energy storage [71]. While carbonates have lower storage temperatures than chlorides, their reduced corrosiveness to metals makes them advantageous in certain applications [72].

Molten salt tanks used for long-duration storage experience heat loss rates of approximately 5–8% per month [69]. The primary technical challenges for molten salt energy storage include the long-term thermal stability of the salts and their corrosiveness to metal materials. Addressing these issues through improved salt formulations, optimized system designs, and the development of corrosion-resistant materials could enhance the viability of molten salt energy storage in future energy systems.

Seasonal Self-Insulating Gravel Thermal Storage System:

ATES, TTES, PTES, and BTES represent the four major technologies currently employed for long-duration thermal storage, enabling large-scale, cross-seasonal thermal energy storage. However, these systems have notable limitations. By integrating the advantages of these technologies, thermal storage's role in LDES could be significantly enhanced. This need has led to the emergence of the seasonal self-insulating gravel thermal storage system.

As shown in Figure 6, this system combines the strengths of the aforementioned technologies by excavating storage pits below the surface and using gravel as the storage material. Concentrating solar collectors are employed as the heat source. Heat is extracted using heat pumps through boreholes. The system takes advantage of gravel's self-insulating effect: under large-scale storage conditions, the outer layers of gravel in the pit cool faster than the inner layers due to their low thermal conductivity. This creates a self-insulating layer, effectively reducing heat loss and enabling low-loss, seasonal long-duration heat storage. Furthermore, with concentrating solar collectors, the storage temperature can be raised to 500 °C, enhancing storage capacity.

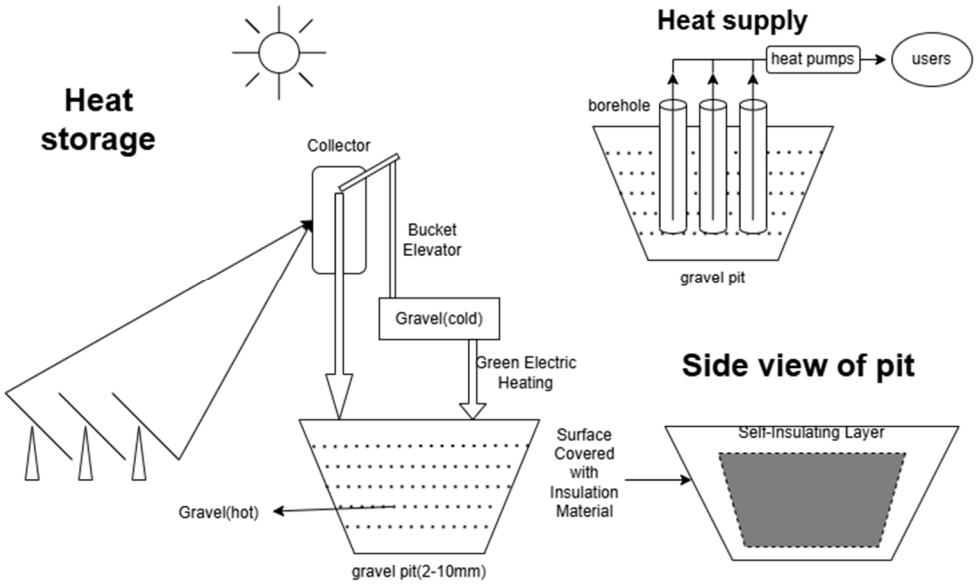


Figure 6. Seasonal self-insulating gravel thermal storage system diagram.

Although the seasonal self-insulating gravel thermal storage system is still in the research stage, its integrated technological advantages and exceptional storage performance provide a new direction for the development of long-duration thermal storage technologies, demonstrating significant potential.

A comparison of different sensible heat storage technologies is shown in Table 3.

Table 3. Comparison of different sensible heat storage technologies.

Thermal Energy Storage Technology	Advantages	Disadvantages
ATES	Low cost Large heat storage capacity	Limited by geographical conditions Risk of groundwater contamination
TTES/PTES	Less dependent on geological conditions Can use liquid–solid mixed materials	Requires larger storage space High tank construction costs
BTES	High storage temperature (~90 °C)	High construction costs Low initial storage efficiency
Molten salt energy storage	High storage temperature (565 °C), large capacity	Poor long-term thermal stability of molten salts Corrosiveness of molten salts to metals
Seasonal self-insulating gravel thermal storage system	High storage temperature (500 °C) Self-insulation with low-cost materials Large heat storage capacity	Still in the research stage

5.1.3. Outlook

The development of sensible heat storage technologies still faces numerous challenges, including limitations related to geological conditions, the stability of high-temperature materials, system heat losses, and construction costs. However, with innovations in material development, system optimization, and technology integration, the future prospects for sensible heat storage remain promising.

On one hand, future research should focus on improving the thermal conductivity and stability of storage media, as well as developing high-performance insulation materials and corrosion-resistant materials. On the other hand, the introduction of intelligent control systems will further optimize storage processes and enhance energy utilization efficiency.

Additionally, the exploration of new forms of thermal storage, such as the seasonal self-insulating gravel thermal storage system, which integrates the advantages of multiple sensible heat storage technologies, offers novel solutions for long-duration heat storage.

As research progresses and applications expand, sensible heat storage technologies are expected to play a greater role in the global energy transition, becoming a critical pillar supporting the large-scale adoption of renewable energy.

5.2. Latent Heat Storage

5.2.1. Overview

In latent heat storage systems, materials used to store thermal energy through phase change are known as phase change materials (PCMs). The phase change process for storing heat can occur in three forms: solid–liquid, liquid–gas, or solid–solid [72]. PCMs are the core of latent heat storage systems, providing efficient heat transfer during energy storage and release.

5.2.2. Types and Technical Characteristics

Based on their chemical composition and phase change properties, PCMs can be divided into three categories: organic PCMs, inorganic PCMs, and eutectic PCMs.

Organic PCMs, represented by paraffins and fatty acids, offer advantages such as good chemical stability, resistance to phase separation, and high reusability. However, they have poor thermal conductivity and relatively low latent heat of fusion [73]. Inorganic PCMs, including salt hydrates and metallic materials, exhibit higher latent heat of fusion compared to organic PCMs. However, they are prone to supercooling and phase separation, resulting in poor long-term stability [73]. Eutectic PCMs, composed of two or more components, combine the advantages of both organic and inorganic PCMs. They can be tailored to specific requirements and are currently mostly in the experimental and research stages [73].

Overall, organic PCMs and eutectic PCMs are more suitable for long-duration latent heat storage applications due to their balanced properties.

5.2.3. Outlook

Latent heat storage has several advantages in thermal energy storage. Compared to sensible heat storage, latent heat storage can store more heat within the same volume, offering higher energy density. During the storage process, temperature fluctuations are minimal, which contributes to system stability. Additionally, its compact design makes latent heat storage particularly suitable for space-constrained, LDES scenarios.

Despite these advantages, latent heat storage faces several challenges. PCMs generally have poor thermal conductivity, limiting the efficiency of heat storage and release. Supercooling and phase separation can occur during the storage process, which may affect system reliability in long-duration applications. The high cost of high-performance PCMs also restricts their large-scale adoption.

Future research should focus on improving thermal conductivity, developing higher-performance composite materials, and optimizing technologies to reduce overall system costs. These advancements will help promote the widespread application of latent heat storage technologies.

5.3. Thermochemical Heat Storage

5.3.1. Overview

Thermochemical heat storage (THS) stores thermal energy through chemical bonds. Compared to sensible and latent heat storage, THS offers significant advantages, drawing widespread attention. Its energy storage density is approximately 8–10 times higher than that of sensible heat storage and twice that of latent heat storage [74]. Additionally, THS experiences virtually no energy loss during long-duration storage. Since thermal energy is stored in the form of chemical bonds, it remains stable over extended periods as long as no chemical reaction occurs [75]. These characteristics make THS a prominent area of research and application in long-duration heat storage.

5.3.2. Types and Technical Characteristics

THS systems can be classified based on their storage mechanisms into solely chemical or thermochemical reaction storage. Chemical storage systems rely on reversible reactions between two chemical substances, typically involving high reaction enthalpy, leading to efficient energy storage. One of the most commonly used reaction pairs is $Ca(OH)_2/CaO$, which offers advantages such as low cost, high energy storage density, good reversibility, stable cycling, non-toxicity, and fast reaction kinetics. The reaction equation is as follows (4).

$$Ca(OH)_2(s) \leftrightarrow CaO(s) + H_2O(g)\ \Delta H = 104.4 kJ/mol \quad (4)$$

During the heat storage process, $Ca(OH)_2$ absorbs heat and decomposes into CaO and water vapor, storing energy in the form of chemical energy. During the heat release process, CaO reacts with water vapor to regenerate $Ca(OH)_2$, releasing the stored chemical energy as heat. At atmospheric pressure, the heat storage temperature of $Ca(OH)_2$ ranges from 400 °C to 600 °C, and the release temperature spans from 25 °C to approximately 500 °C, providing flexibility in adjusting the power of heat storage and release [76].

Improving the heat storage performance of the material can achieve higher heat storage capacity, as shown in Figure 7. Various material enhancement strategies have been explored to address the key challenges of $Ca(OH)_2$, including low thermal conductivity, high decomposition temperature, and material degradation over multiple cycles. These strategies can be classified into powder material modifications and granulated material improvements.

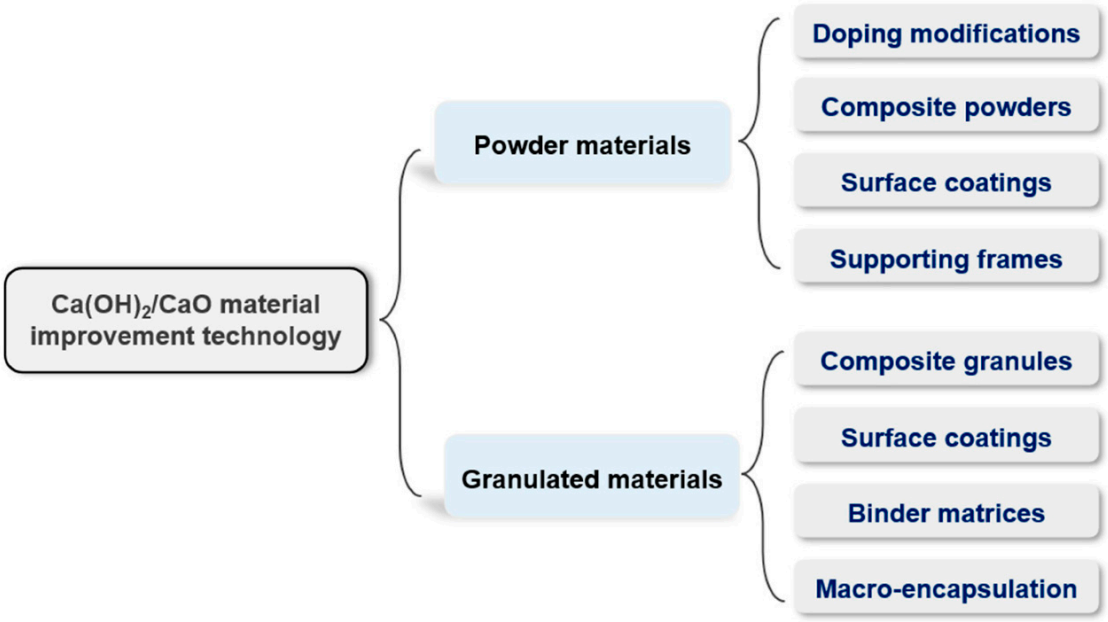


Figure 7. Classification of material performance enhancement techniques [76].

Powder material modifications include doping modifications, where elements such as Al, Ni, Zn, and Li are introduced to lower the dehydration temperature and enhance reaction kinetics [77]. Additionally, composite powders incorporating expanded graphite and nano-SiO_2 improve thermal conductivity and prevent agglomeration, thereby ensuring stable long-term cycling performance [78]. Surface coatings with nano-SiO_2 can further mitigate agglomeration and enhance flow properties, while supporting frames composed of ceramic or silicon carbide improve structural stability and hydration performance [79].

In contrast, granulated material improvements focus on enhancing mechanical stability and cycling durability. Composite granules containing calcium titanate ($CaTiO_3$) provide structural reinforcement, mitigating volume changes during hydration-dehydration cycles [80]. Surface coatings, such as alumina or silica, help maintain material integrity over repeated cycles [81]. Binder matrices using sodium silicate improve pellet strength, reducing fragmentation and enhancing reversibility [82]. Lastly, macro-encapsulation techniques utilizing semi-permeable shells minimize material degradation and moisture exposure, ensuring long-term efficiency [83].

Thermochemical reaction storage systems store thermal energy through chemical bonds formed during the adsorption of chemical substances. These adsorption reactions generally occur at lower temperatures and require low activation energy, making them particularly suitable for low-temperature heat storage applications [74,84].

5.3.3. Outlook

Despite its great potential, THS development faces several technical challenges. The efficiency of heat and mass transfer within the system is often low, preventing the complete reaction of the reactants and reducing storage efficiency. Additionally, the high operational energy consumption further limits the system's overall energy utilization. Another key issue is the poor cycling stability of thermal storage materials, with significant performance degradation over repeated storage cycles.

Future research should focus on developing high-performance composite materials to optimize storage density, improve heat transfer efficiency, and enhance cycling stability. Such advancements are expected to facilitate the widespread application of THS in long-duration heat storage systems.

6. Discussion and Conclusions

This paper provides a detailed summary of the technical principles, performance characteristics, and application scenarios of four major LDES technologies: mechanical energy storage, chemical energy storage, electrochemical energy storage, and thermal energy storage. A comparison of the storage media, storage durations, and round-trip efficiencies of these LDES technologies is presented in Table 4.

Table 4. Comparison of LDES technologies.

LDES Technologies		Storage Medium	Storage Form	Storage Duration	Storage Efficiency (or Self-Loss Rate)
Mechanical energy storage	PHES	Water	Gravitational potential energy	Several hours to several days	~80%
	CAES	Air	Elastic potential energy	Several hours to several days	55–75%
Chemical energy storage	Hydrogen energy storage	Hydrogen	Chemical energy	Several months or longer	Liquid hydro-gen evaporation loss ~3%/day
	Synthetic Fuel Storage	Fuel	Chemical energy	Several months or longer	—
Electrochemical energy storage	LIB	Lithium salts	Chemical energy	Several hours to several days	~97%, self-discharge rate as low as 0.03%
	RFB	Electrolyte	Chemical energy	Several days	65–80%
Thermal energy storage	ATES	Water	Thermal energy	Seasonal (several months)	67.5–87%
	TTES	Water/water-Gravel	Thermal energy	Seasonal (several months)	45–65%
	BTES	Subsurface	Thermal energy	Seasonal (several months)	40–60%
	Molten salt energy storage	Molten salts	Thermal energy	Seasonal (several months)	5–8%/month
	Gravel thermal storage system	Gravel	Thermal energy	Seasonal (several months)	—
	LHS	Chemical materials	Chemical energy	Several months or longer	—
	THS	Chemical materials	Chemical energy	Several months or longer	—

PHES and CAES, as traditional LDES technologies, offer significant advantages in terms of cost and scalability, making them highly suitable for large-scale integration of renewable energy. In chemical energy storage, hydrogen and synthetic fuels demonstrate exceptional flexibility in LDES applications, with the stored energy applicable across diverse scenarios and even capable of replacing fossil fuels.

Electrochemical energy storage technologies, particularly LIBs and RFBs, have emerged as innovative solutions in recent years. Advances in research have resulted in declining costs and improved safety for electrochemical energy storage, which exhibits the highest energy density among LDES technologies. These attributes make it highly promising for large-scale renewable energy integration. Furthermore, electrochemical energy storage can be effectively combined with other LDES technologies to further enhance storage capacity.

Thermal energy storage, utilizing sensible heat, latent heat, and thermochemical heat storage technologies, enables large-scale heat storage. It provides essential support for cross-seasonal energy dispatch, alleviating the mismatch between heat energy supply and demand.

LDES technologies are indispensable for supporting large-scale renewable energy integration and ensuring the flexibility and reliability of energy systems. By fostering technological innovation, optimizing resource allocation, and promoting interdisciplinary collaboration, the future of LDES will achieve not only technical breakthroughs but also play a pivotal role in advancing the global energy system toward low-carbon transformation and achieving sustainable development goals.

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